IMPERIAL COLLEGE LONDON

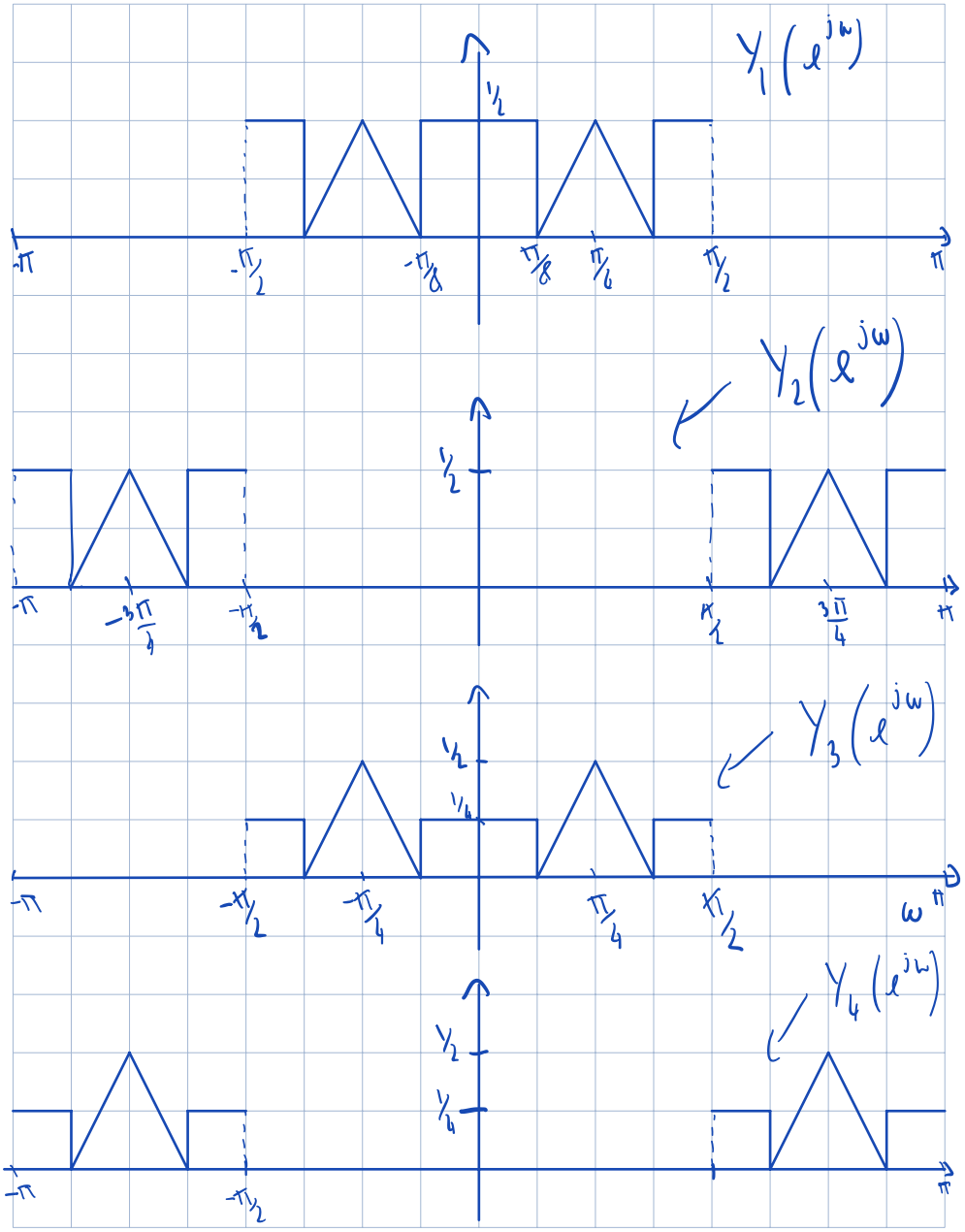
DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

EXAMINATIONS 2025

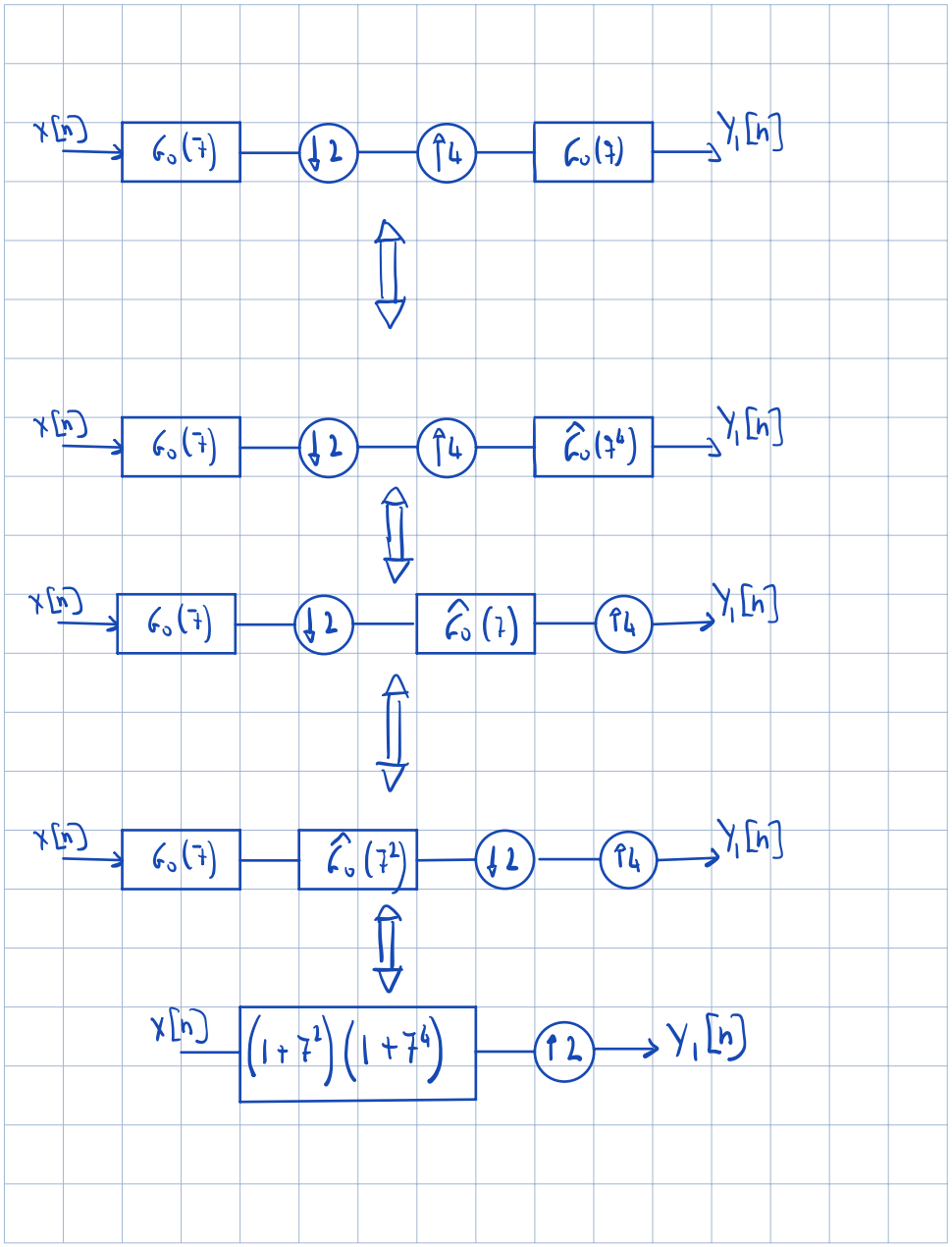
WAVELETS AND APPLICATIONS SOLUTIONS

SOLUTIONS

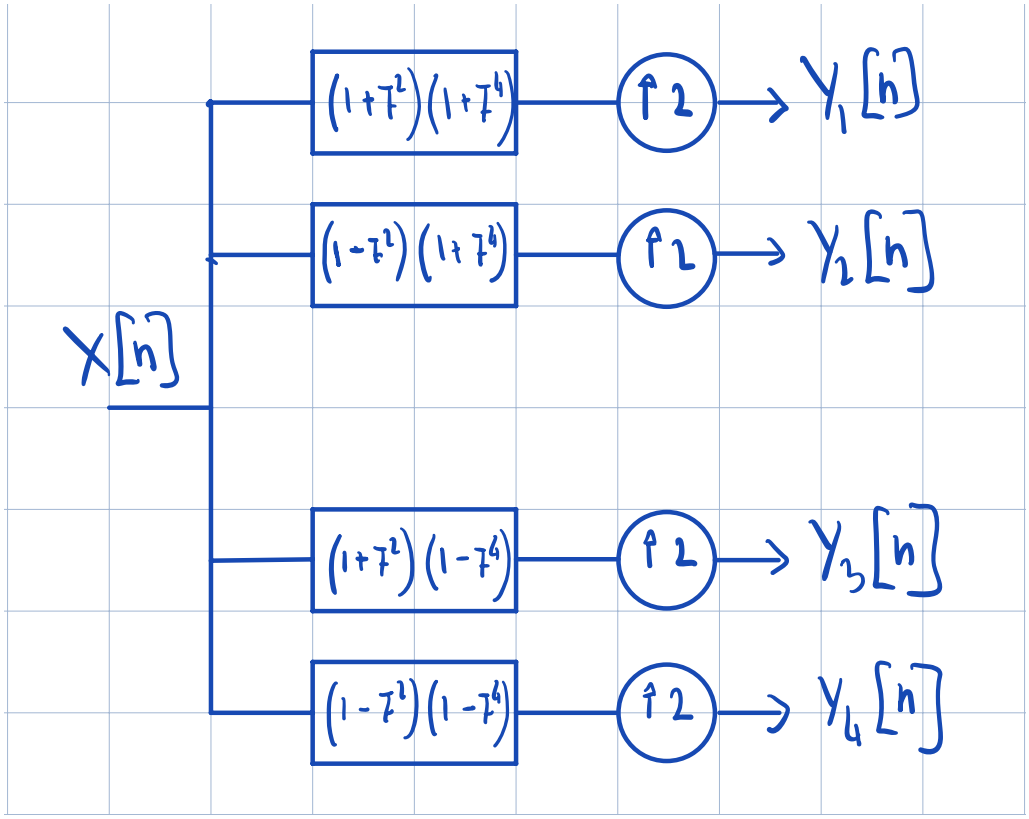
1. (a) The four spectra are given below:



(b) i. We define $\hat{G}_0(z) = (1+z)$ and observe that $G_0(z) = \hat{G}_0(z^4)$. Based on this observation, we have the following equivalences for the upper branch:



By using the same approach, we arrive at the following four equivalent filters:



ii. The filter $G_1(z)$ has one zero at $\omega = 0$, this means that it annihilates constant polynomials. For this reason only $y_1[n] \neq 0$.

iii. You need a filter with two zeros at $\omega = 0$ to annihilate linear polynomials. We note that only the equivalent filter $G_4(z) = (1 - z^2)(1 - z^4)$ satisfies this property. This means that the only zero sequence is $y_4[n]$.

iv. The easiest option is to pick $G_1(z) = (1 - z^4)(1 - z^4)$. In this way the new equivalent filter $G_4(z) = (1 - z^2)^2(1 - z^4)^2$ has four zeros at $\omega = 0$ and so can annihilate polynomials up to degree three. The other equivalent filters have at most two zeros at $\omega = 0$ and so cannot annihilate n^2.

2. (a)

$$\hat{X}(z) = \frac{G_0(z)}{2}\left[H_0(z)X(z) + H_0(-z)X(-z)\right] + \frac{G_1(z)}{2}\left[H_1(z)X(z) + H_1(-z)X(-z)\right].$$

Therefore PR condition is:

$$G_0(z)H_0(z) + G_1(z)H_1(z) = 2$$

and

$$G_0(z)H_0(-z) + G_1(z)H_1(-z) = 0.$$

(b) i. We have:

$$G_0(z) = H_0(z^{-1}) = \frac{1}{\sqrt{2}}(1 + z^{-3}),$$

[3/7]

$$G_1(z) = -z^{-1}G_0(-z^{-1}) = \frac{1}{\sqrt{2}}z^{-1}(z^3 - 1)$$

and

$$H_1(z) = G_1(z^{-1}) = \frac{1}{\sqrt{2}}z(z^{-3} - 1).$$

[7/7]

ii. Given that

$$P(z) = G_0(z)H_0(z) = \left(\frac{z^3}{2} + 1 + \frac{z^{-3}}{2}\right)$$

and

$$H_0(z) = (z - 1 + z^{-1}),$$

we realise that $G_0(z) = \frac{1}{2}(z - 1 + z^{-1})(1 + z)(1 + z^{-1})$.

[3/7]

The other two filters are: $G_1(z) = z^{-1}H_0(-z)$ and $H_1(z) = zG_0(-z)$.

[7/7]

(c) i. In this case we have that

$$\hat{X}(z) = \left[H_0(z)G_0(z) + H_1(z)G_1(z)\right]X(z).$$

Since $\{H_0(z), H_1(z), G_0(z), G_1(z)\}$ form an orthogonal filter bank, it follows that $H_0(z)G_0(z) + H_1(z)G_1(z) = 2$ and that $\hat{x}[n] = 2x[n]$.

ii. PR condition in this case reduces to

$$H_0(z)G_0(z) + H_1(z)G_1(z) = 1.$$

This means that in this specific case we require $H_1(z)G_1(z) = 1 - H_0(z)G_0(z) = -2z^{-1} - 4 - 2z$. We note that $P(z) = H_1(z)G_1(z)$ has two real roots and can be written as follows: $P(z) = -2(z + 1)(z^{-1} + 1)$. Therefore using spectral factorization we can find several solutions including:

$$H_1(z) = -2 \text{ and } G_1(z) = (z + 1)(z^{-1} + 1)$$

or

$$H_1(z) = z^{-1} + 1 \text{ and } G_1(z) = -2(z + 1).$$

3. (a) Since the functions $\varphi(t)$, $\psi(t)$, $\sqrt{2}\psi(2t)$ and $\sqrt{2}\psi(2t-1)$ are orthonormal, we don't need to find the dual basis: $\tilde{\varphi}_i(t) = \varphi_i(t)$. [1/5]

Consequently

$$c_1 = \langle f(t), \varphi(t) \rangle = \pi \int_0^{1/2} \sin(2\pi t) dt = 1$$

$$c_2 = \langle f(t), \psi(t) \rangle = \pi \int_0^{1/2} \sin(2\pi t) dt = 1$$

$$c_3 = \langle f(t), \sqrt{2}\psi(2t) \rangle = \sqrt{2}\pi \left(\int_0^{1/4} \sin(2\pi t) dt - \int_{1/4}^{1/2} \sin(2\pi t) dt \right) = 0$$

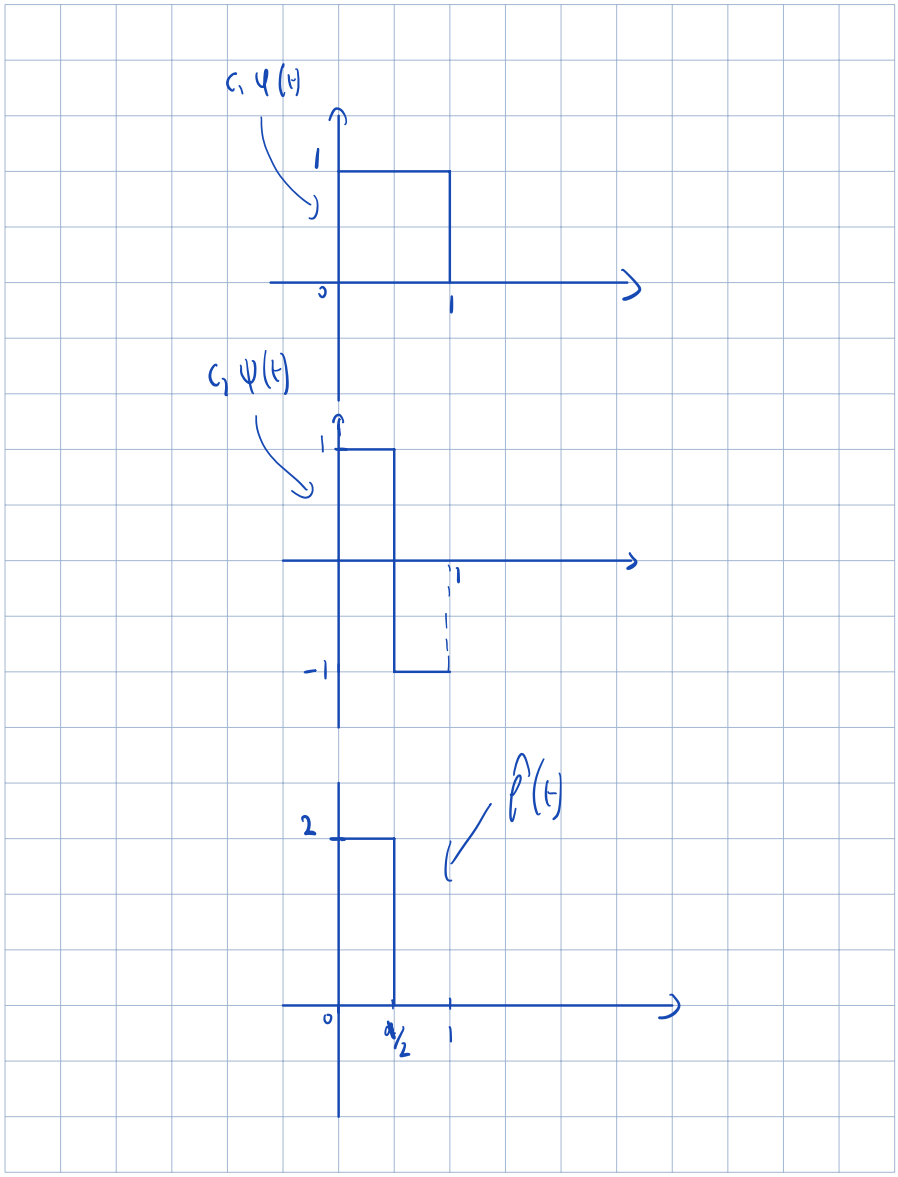
$$c_4 = \langle f(t), \sqrt{2}\psi(2t-1) \rangle = 0$$

[5/5]

(b) In this case

$$\hat{f}(t) = \varphi(t) + \psi(t)$$

and the solution is sketched below:



(c)

$$\|\epsilon(t)\|^2 = \int_0^{1/2} (\pi \sin(2\pi t) - 2)^2 dt = \pi^2/4 - 2$$

(d) Using Parseval identity we have that

$$\|\hat{f}(t)\|^2 = \sum_{i=1}^{4} c_i^2 = 2.$$

Moreover,

$$\|f(t)\|^2 = \pi^2 \int_0^{1/2} \sin(2\pi t)^2 dt = \pi^2/4.$$

Therefore

$$\|\epsilon(t)\|^2 + \|\hat{f}(t)\|^2 = \pi^2/4 - 2 + 2 = \|f(t)\|^2.$$

(e) For the Haar expansion with $f(t) = \varphi(t)$, the Haar wavelet coefficients are:

$$a_{m,n} = \int_0^1 \psi_{m,n}(t) dt = \begin{cases} 2^{-m/2} & m \geq 1, n = 0 \\ \\ 0 & \text{otherwise} \end{cases}$$

4. (a) Using the definition of the Fourier transform, we have

$$\hat{\varphi}(\omega) = \sqrt{2}\sum_n g_0[n]\int_{-\infty}^{\infty}\varphi(2t-n)e^{-j\omega t}dt.$$

The change of variables $x = 2t - n$ yields

$$\hat{\varphi}(\omega) = \frac{\sqrt{2}}{2}\sum_n g_0[n]e^{-j\omega n/2}\int_{-\infty}^{\infty}\varphi(x)e^{-j\omega x/2}dx = \frac{1}{\sqrt{2}}G_0(e^{j\omega/2})\hat{\varphi}(\omega/2),$$

where the last equality follows from the fact that $\sum_n g_0[n]e^{-j\omega n/2}$ is the discrete-time Fourier transform of $g_0[n]$ at $\omega/2$ and that

$$\int_{-\infty}^{\infty}\varphi(x)e^{-j\omega x/2}dx$$

is the Fourier transform of $\varphi(t)$ at $\omega/2$.

(b) We need to show that for any arbitrary small $\epsilon > 0$ we can find a δ such that, if $|x - t| < \delta$ then $|f(t) - f(x)| < \epsilon$. We use the expression of the inverse Fourier transform

$$f(t) = \frac{1}{2\pi}\int \hat{f}(\omega)e^{j\omega t}d\omega$$

and note that

$$\begin{aligned}
|f(t) - f(x)| &= \tfrac{1}{2\pi}\left|\int \hat{f}(\omega)(e^{j\omega t} - e^{\omega x})d\omega\right| \\
&\leq \tfrac{1}{2\pi}\int |\hat{f}(\omega)|\left|(e^{j\omega t} - e^{\omega x})\right|d\omega \\
&\leq \tfrac{c}{2\pi}|t-x|\int |\hat{f}(\omega)|d\omega \\
&\leq \tfrac{c'}{2\pi}|t-x|,
\end{aligned}$$

where in the last two inequalities we have used the fact that $\left|(e^{j\omega t} - e^{\omega x})\right| \leq c|x-t|$ when $x - t$ is small and the hypothesis that

$$\int |\hat{f}(\omega)|d\omega < +\infty$$

respectively. In conclusion we have shown that:

$$|f(t) - f(x)| \leq c''|t - x|$$

which proves continuity.

(c) i. $G_0(z) = \sqrt{2}\left(\frac{1+z^{-1}}{2}\right)^4$, therefore $G_0(e^{j\pi}) = 0$ and $G_0(1) = \sqrt{2}$, so the necessary conditions for convergence are satisfied.

ii. In this case, we can write $M_0(\omega)$ as follows

$$M_0(\omega) = \left(\frac{1 + e^{-j\omega}}{2}\right)^N R(\omega),$$

where $R(\omega) = 1$ and $N = 4$. This means that $B = \max_\omega R(\omega) = 1$ and since $B < 2^{N-3}$ we know that the sufficient condition for $\varphi(t)$ to be continuous with its derivatives up to order two is satisfied. In fact, the iteration converges to the cubic spline.

iii. $G_0(z)$ has four zeros at $\omega = \pi$, therefore the analysis wavelet has four vanishing moments.